

A Counterexample to Conjecture 00000007780: Four Common Preperiodic Points of $z^2 - 1$ and z^2

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Abstract

Conjecture 00000007780 claims that a quadratic polynomial f and a power map z^d have at most $2\nu(d)$ common preperiodic points on $\mathbb{P}^1(\overline{\mathbb{Q}})$, where $\nu(d)$ is the number of prime factors of d , and at most 3 if f is neither Chebyshev nor monomial. For $f(z) = z^2 - 1$ and $d = 2$ the points $0, 1, -1, \infty$ are preperiodic for both maps. Since $4 > 2 = 2\nu(2)$ and $z^2 - 1$ is neither Chebyshev nor monomial, both claims fail. The refutation is checked in Lean 4 against Mathlib.

1 The conjecture as stated

The original text of conjecture 00000007780 reads:

Definition: Common preperiodic sets of mixed power maps: $S(d_1, d_2) = \text{Per}(z^{d_1}) \cap \text{Per}(z^{d_2})$ consists of roots of unity. Conjecture: For any quadratic polynomial f together with a power map z^d on $\mathbb{P}^1(\overline{\mathbb{Q}})$, the common preperiodic set has cardinality uniformly bounded by $2v(d)$ with $v(d)$ the prime factor count of d ; and when f is neither Chebyshev nor monomial the set contains only fixed-point-type points, at most 3, decidable by uniform distribution of the fixed-point multipliers of f .

The Chinese version in `conjectures/00000007780.md` states the same two claims.

2 Reading of the statement

A polynomial map g of positive degree acts on $\mathbb{P}^1(\overline{\mathbb{Q}}) = \overline{\mathbb{Q}} \cup \{\infty\}$, fixing ∞ . A point x is *preperiodic* for g if its forward orbit $\{g^{\circ k}(x) : k \geq 0\}$ is finite, equivalently if some iterate $g^{\circ m}(x)$ is periodic. Write $\text{Prep}(g)$ for the set of preperiodic points.

For a quadratic polynomial f and an integer $d \geq 2$ the *common preperiodic set* is $\text{Prep}(f) \cap \text{Prep}(z^d) \subseteq \mathbb{P}^1(\overline{\mathbb{Q}})$. A quadratic polynomial is *monomial* if it is affinely conjugate to z^2 , and *Chebyshev* if it is affinely conjugate to $T_2(z) = 2z^2 - 1$ or to $-T_2$. Conjugacy is by $\varphi(z) = \alpha z + \beta$ with $\alpha \neq 0$, meaning $\varphi \circ f = g \circ \varphi$. These are the broadest readings: a literal monomial az^2 or a literal $\pm T_2$ is in particular conjugate to z^2 or T_2 .

The two claims are:

(C1) for every quadratic polynomial f and every $d \geq 2$, $\#(\text{Prep}(f) \cap \text{Prep}(z^d)) \leq 2\nu(d)$;

(C2) if f is neither Chebyshev nor monomial, then $\#(\text{Prep}(f) \cap \text{Prep}(z^d)) \leq 3$ for every $d \geq 2$.

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The “prime factor count” $\nu(d)$ may be read with or without multiplicity; for $d = 2$ both readings give $\nu(2) = 1$. Restricting to $d \geq 2$ only weakens (C1) and (C2), so refuting the restricted claims is stronger. In (C2) we refute only the bound “at most 3”; the other phrases of (C2) are not needed.

3 Result

Theorem 1. *Let $f(z) = z^2 - 1$ and $d = 2$. Then $\{0, 1, -1, \infty\} \subseteq \text{Prep}(f) \cap \text{Prep}(z^2)$, so the common preperiodic set has at least 4 elements. Moreover f is neither monomial nor Chebyshev. Hence both (C1) (as $4 > 2 = 2\nu(2)$) and (C2) (as $4 > 3$) fail, and Conjecture 00000007780 is false.*

4 Proof

Preperiodicity. Under $f(z) = z^2 - 1$ we have $f(0) = -1$, $f(-1) = 0$ and $f(1) = 0$. So 0 and -1 are periodic of period 2, and 1 is strictly preperiodic. Under $z \mapsto z^2$ the points 0 and 1 are fixed and $-1 \mapsto 1$. Both maps fix ∞ . Thus 0, 1, -1 , ∞ are preperiodic for both maps, and they are pairwise distinct since $\overline{\mathbb{Q}}$ has characteristic 0.

f is not monomial. Suppose $\alpha f(z) + \beta = (\alpha z + \beta)^2$ for all z , with $\alpha \neq 0$. At $z = 1$ and $z = -1$ (where $f = 0$) this gives $\beta = (\alpha + \beta)^2$ and $\beta = (\beta - \alpha)^2$. Subtracting yields $4\alpha\beta = 0$, so $\beta = 0$. Then $0 = \alpha^2$ at $z = 1$, contradicting $\alpha \neq 0$.

f is not Chebyshev. Suppose $\alpha f(z) + \beta = \varepsilon(2(\alpha z + \beta)^2 - 1)$ for all z , with $\alpha \neq 0$ and $\varepsilon \in \{1, -1\}$. At $z = \pm 1$ we get $\beta = \varepsilon(2(\alpha + \beta)^2 - 1) = \varepsilon(2(\beta - \alpha)^2 - 1)$, hence $8\varepsilon\alpha\beta = 0$ and $\beta = 0$. At $z = 0$ we get $-\alpha = -\varepsilon$, so $\alpha = \varepsilon$. At $z = 1$ we then get $0 = \varepsilon(2\alpha^2 - 1) = \varepsilon$, a contradiction.

5 Formalization

The Lean 4 project in `lean/` (file `lean/Submission/Basic.lean`) formalizes the definitions above and proves Theorem 1 against Mathlib.

- $\overline{\mathbb{Q}}$ is `AlgebraicClosure` of $\mathbb{Q}$, and $\mathbb{P}^1(\overline{\mathbb{Q}})$ is `Option Qbar` with `none` as ∞ . A polynomial map acts by `Option.map`, which fixes ∞ .
- `IsPreperiodic F x` means that some iterate $F^{\text{om}}(x)$ is a periodic point in the sense of Mathlib’s `Function.IsPeriodicPt` (with positive period).
- `quad a b c` is $z \mapsto az^2 + bz + c$, `powMap d` is $z \mapsto z^d$, and `commonPrep a b c d` is the common preperiodic set. `IsMonomial` and `IsChebyshev` are the conjugacy conditions above.
- `Claim1` is (C1) with $\nu = \omega$ (`ArithmeticFunction.cardDistinctFactors`); `Claim1’` is (C1) with $\nu = \Omega$ (`ArithmeticFunction.cardFactors`); `Claim2` is the bound in (C2). Cardinalities are `Set.encard`, so infinite sets are handled correctly.
- `subset_commonPrep` and `four_le_encard` give the four common points; `not_isMonomial` and `not_isChebyshev` are the two computations above.
- `not_claim1`, `not_claim1’` and `not_claim2` are the refutations, and `conjecture_00000007780_false` states $\neg(\text{Claim1} \vee \text{Claim1’}) \wedge \neg \text{Claim2}$.

The toolchain is `leanprover/lean4:v4.33.1`, Mathlib is pinned to `v4.33.1`, and `lake build` succeeds. The project contains no `sorry`, no `admit` and no `native_decide`, and introduces no axioms: `#print axioms conjecture_00000007780_false` reports exactly `propext`, `Classical.choice` and `Quot.sound`.

6 Remarks

Remark 2. None of $0, 1, -1$ is a fixed point of f (the fixed points of $z^2 - 1$ are $(1 \pm \sqrt{5})/2$), so the description “only fixed-point-type points” in (C2) also fails under the natural reading. We formalize only the count.

Remark 3. For monomial f the bound (C1) fails badly: $\text{Prep}(z^2) \cap \text{Prep}(z^d)$ contains every root of unity, hence is infinite. Our example is not monomial, so it refutes (C1) even for readings that tacitly exclude the monomial case.